

Step	Step Description	Agents Involved	Tool Calls and Parameters	Reason
1	User asks: Where are houses in Exeter that were built before 1960?	human, host_agent	Message from human to host_agent	User initiates the query.
2	Plan steps to solve the geospatial query	planning_agent	send_message to planning_agent with task_description	planning_agent is chosen to break down the query into actionable steps, as per its system prompt.
3	Planning agent decomposes the query	planning_agent	Output steps: ["Find Exeter 1 area", "Find houses in Exeter as many search results", "Apply condition for houses built before 1960"]	Steps are based on the planning_agent's prompt: identify area, search for entities, apply conditions.
4	Find the area polygon for Exeter	host_agent, named_area_agent	send_message to named_area with task_description: 'Find the area polygon for Exeter.'	named_area_agent is chosen because the prompt specifies it provides polygons for named areas like cities.
5	Retrieve polygons for Exeter	named_area_agent	call_os_ngd with polygon_or_point=True, filename='exeter_area_polygon'	named_area_agent uses the OS NGD tool to get all named area polygons, as per its capabilities.
6	Filter to the specific Exeter polygon	host_agent, data_analysis_agent	send_message to data_analysis_agent with task_description: Filter 'exeter_area_polygon' for Exeter, save as 'filtered_exeter_area_polygon'	Data may contain multiple polygons; filtering is needed to isolate the city of Exeter.
7	Generate metadata for 'exeter_area_polygon'	data_analysis_agent	generate_metadata_for_artifacts with artifact_names=['exeter_area_polygon']	Metadata is generated to understand the columns and structure for accurate filtering.
8	Filter code execution	coding_agent, code_executor	code_executor with code to filter for name1_text == 'Exeter' and description in ['City', 'Settlement', 'Part of Settlement']	Code uses 'name1_text' and 'description' columns as per metadata to accurately select the Exeter city polygon.
9	Save filtered Exeter polygon	data_analysis_agent	Output: 'filtered_exeter_area_polygon'	The result is a GeoDataFrame with only the Exeter city polygon, as required for further spatial queries.
10	Find houses in Exeter built before 1960	host_agent, buildings_agent	send_message to buildings with task_description: Find houses in 'filtered_exeter_area_polygon' built before 1960	buildings_agent is chosen because it queries the buildings database; bbox is set to the filtered Exeter polygon.
11	Retrieve buildings in Exeter area	buildings_agent	call_os_ngd with bbox='filtered_exeter_area_polygon', filename='exeter_buildings_before_1960'	The buildings_agent uses the OS NGD tool to get all buildings in the Exeter area polygon.
12	Filter for houses built before 1960	host_agent, data_analysis_agent	send_message to data_analysis_agent with task_description: Filter 'buildings_exeter_buildings_before_1960' for houses built before 1960 using buildingage_period, buildinguse_addresscount_residential > 0, buildinguse_addresscount_total = 1	The prompt for buildings_agent specifies that a house is defined by these address count columns; buildingage_period is used for pre-1960 buildings as per metadata.
13	Generate metadata for buildings artifact	data_analysis_agent	generate_metadata_for_artifacts with artifact_names=['buildings_exeter_buildings_before_1960']	Metadata is generated to confirm column names and valid values for filtering.
14	Filter code execution for houses before 1960	coding_agent, code_executor	code_executor with code to filter for buildingage_period in ['Pre-1837', '1837-1869', '1870-1918', '1919-1944', '1945-1959', 'Pre-1919'], buildinguse_addresscount_residential > 0, buildinguse_addresscount_total = 1	Code uses the correct columns and values as per metadata and system prompt for houses and construction period.
15	Save filtered houses artifact	data_analysis_agent	Output: 'filtered_houses_before_1960_exeter'	The result is a GeoDataFrame of houses in Exeter built before 1960, with exactly one residential address.
16	Generate metadata for all artifacts	host_agent	generate_metadata_for_all_artifacts	Ensures all artifacts are described for downstream use, especially for plotting.
17	Plot houses within Exeter area	host_agent, plotting_agent	send_message to plotting_agent with task_description: Plot 'filtered_houses_before_1960_exeter' within 'filtered_exeter_area_polygon'	plotting_agent is chosen to visualize the spatial distribution of the filtered houses.
18	Generate metadata for plotting	plotting_agent	generate_metadata_for_artifacts with artifact_names=['filtered_houses_before_1960_exeter', 'filtered_exeter_area_polygon']	Metadata is needed to confirm geometry columns and CRS for plotting.
19	Plotting code execution (first attempt, error)	coding_agent, code_executor	code_executor with code to plot houses and area using folium, but error due to non-serializable Timestamp columns	The error occurs because folium cannot serialize pandas Timestamps; this is a common issue when plotting GeoDataFrames with datetime columns.
20	Plotting code execution (second attempt, success)	coding_agent, code_executor	code_executor with revised code: converts datetime columns to string before plotting	The code is revised to clean datetime columns, ensuring folium can serialize the data for plotting.
21	Save and summarize plot	plotting_agent	Output: houses_within_exeter_area.html, summary of first 5 houses	The map is saved as an HTML file, and a summary of the first 5 houses is provided for user context.
22	Final output to user	host_agent	Output: Summary of results and map artifact	host_agent summarizes the findings and offers the map for download or further queries.

Overall Summary: - Databases used: Named area (for Exeter polygon), Buildings (for houses in Exeter), and plotting tools for visualization. - Filters/code: Filtering for the Exeter city polygon used 'name1_text' and 'description' columns. Filtering for houses used 'buildingage_period' for pre-1960, and 'buildinguse_addresscount_residential' > 0 and 'buildinguse_addresscount_total' = 1 to define houses, as specified in the buildings_agent prompt and metadata. - The plotting step required cleaning datetime columns to avoid serialization errors with folium. - Final outcome: 19,784 houses in Exeter built before 1960 were identified and plotted on a map, with the map saved as an HTML artifact for user access. - Overall reasoning: Each agent and tool was chosen based on its system prompt and the requirements of the query. Metadata was used at each step to ensure correct column usage and filtering logic, and errors in plotting were resolved by adapting the code to the data's structure.